

Predictors of seizure outcome in newly diagnosed partial epilepsy: memory performance as a prognostic factor

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Received 21 March 1999; accepted 6 June 1999

Abstract

The epilepsy patients whose seizures will prove to be refractory should be identified as early as possible, and thus the need for new prognostic factors of intractable epilepsy is evident. The aim of the study was to investigate predictors of seizure outcome in a multivariate analysis. Neurological, electroencephalography (EEG) and neuropsychological variables were analyzed as potential predictors of epilepsy. Eighty-nine newly diagnosed adult patients with partial epilepsy were, after a prospective 2-year follow-up period, categorized into one of the two groups: patients with satisfactorily controlled epilepsy, and patients with refractory epilepsy. Six variables predicted 2-year seizure outcome: presence of spike focus in EEG, partial complex or mixed seizure type, remote symptomatic etiology, moderately impaired memory performance in immediate recall and in delayed recognition of the word list, and age at the time of diagnosis. The correct seizure outcome could be predicted with the model in 94% of newly diagnosed epilepsy patients. The presence of verbal memory impairment at the time of the diagnosis of partial epilepsy is a significant predictor of seizure outcome and, together with clinical and EEG variables, it predicts seizure outcome in the majority of the patients. Memory performance as a prognostic factor is of most value in patients with risk of refractory epilepsy and when used in a multidisciplinary setting. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: Epilepsy; Outcome; Memory; Prognosis; Cognition

1. Introduction

Up to 80% of patients with newly diagnosed epilepsy will have their seizures well controlled with appropriate antiepileptic drug treatment

(Mattson, 1994). On the other hand, the seizure control remains suboptimal for many patients, and 20–30% of the patients will have chronic drug-resistant epilepsy (Smith et al., 1987; Keränen and Riekkinen, 1988; Collaborative Group for the Study of Epilepsy, 1992; Cockerell et al., 1995). At the time of diagnosis, however, it is usually not possible to predict which patient is going to develop an intractable disorder.

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The ultimate prognosis for epilepsy is probably largely determined by the underlying cause of the seizures, the particular epilepsy syndrome, and whether there are associated neurologic abnormalities (Shinnar and Berg, 1996). Seizures are only one of the many factors that contribute the overall morbidity in patients with epilepsy, but remission of the seizures is the most obvious measure of the course of the disease. The most consistently reported adverse prognostic factors for seizure outcome are the presence of complex partial or mixed seizure type and of remote symptomatic etiology (Elwes and Reynolds, 1991; Collaborative Group for the Study of Epilepsy, 1992; Mattson et al., 1996). On the other hand, contradictory results from the National General Practice Study of Epilepsy (Cockerell et al., 1995) showed that seizure type, etiology and age had little effect on the chances of achieving remission. A large pretreatment number of seizures (> 10) has been shown to be associated with poorer prognosis of subsequent seizure control (Elwes et al., 1984; Sillanpää, 1993; Camfield et al., 1996). Several studies have shown that early treatment response is a good predictor of long-term seizure outcome; especially, the first 2 years have been suggested to be crucial for the long-term prognosis of epilepsy (Elwes et al., 1984, 1988; Collaborative Group for the Study of Epilepsy, 1992; Mattson et al., 1996). However, it is still unclear why some of the patients develop an intractable seizure disorder.

Epilepsy should, however, not be viewed only as a random succession of seizures, but as a dynamic process, which results in both interictal functional abnormalities as well as pathological changes in the brain structure. The prognosis of the epilepsy might be improved by early, effective intervention, and there is a pressing need to be able to identify already at the time of diagnosis those patients whose seizures will prove to be intractable.

Development and refinement of imaging methods and non-invasive measures of interictal brain function, such as functional imaging, will provide clinically useful prognostic information of the epilepsy. Neurophysiological methods have traditionally been used in epilepsy prognosis studies, but measures of cognitive functioning have not so

far been used as a prognostic factor for newly diagnosed epilepsy. Neuropsychologic data have, however, been related to seizure outcome after epilepsy surgery (Dodrill et al., 1986; Perrine et al., 1995), and preoperative IQ score has been shown to be statistically significant predictor of seizure outcome (Chelune et al., 1998). In temporal lobe epilepsy and in evaluation for epilepsy surgery, assessment of memory function is crucial, because of the critical role of medial temporal lobe structures in memory as well as in epileptic process (Jones-Gotman et al., 1993).

The aim of this prospective 2-year follow-up study was to investigate predictors of seizure outcome and to determine the contribution of verbal memory performance to seizure outcome. First, the objective was to find whether patients with memory impairment at the time of diagnosis of partial epilepsy will later develop intractable epilepsy, and second, to evaluate the prognostic value of verbal memory impairment together with clinical prognostic factors for 2-year seizure outcome in newly diagnosed partial epilepsy.

2. Patients and methods

2.1. Patients

Eighty-nine consecutive untreated adult patients, aged between 15 and 63 years, with newly diagnosed partial epilepsy were included in the study. Patients with alcohol-related seizures, current alcohol or other drug abuse, progressive neurological disorders, mental retardation, severe psychiatric problems, or other severe medical disorders were excluded from the study. The pretreatment seizures were classified according the revised International Classification of Epileptic Seizures (Commission on Classification and Terminology of the International League Against Epilepsy, 1981). Subjects underwent a full neurological examination, brain computed tomography and/or magnetic resonance imaging, a comprehensive neuropsychological evaluation and electroencephalography (EEG) before starting antiepileptic drug treatment. On the basis of all available information, the putative etiology of the

epilepsy was determined. Patients with unprovoked epileptic seizures were included in the study. Both patients with remote symptomatic seizures, which follow central nervous system (CNS) insults by more than 1 week, and patients with cryptogenic seizures, which do not have any known associated CNS or other etiological disorder (Hauser and Kurland, 1975), were included in the study. Forty-eight healthy volunteers served as a control group for the neuropsychological evaluation.

After a prospective follow-up period, the patients were divided into two groups according to 2-year seizure outcome: Group I patients had satisfactorily controlled epilepsy (seizure free or only occasional seizures), and Group II patients had refractory seizure disorder despite adequate drug treatment. The seizure disorder was termed refractory if more than one generalized or more than four partial seizures occurred during 1 year. Number of seizure-free patients was also recorded.

2.2. Verbal memory assessment and EEG scoring

Verbal memory performance and general intellectual ability were assessed as a part of a comprehensive neuropsychological evaluation, reported in detail by Kälviäinen et al. (1995).

The Logical Prose Subtest, Story A of the Wechsler Memory Scale (Wechsler, 1945) was used as a measure for logical verbal memory. The immediate verbatim recall score was based on the number of story units immediately recalled. The delayed score was based on the units recalled from the story 90 min later.

The List Learning Test was used to assess learning and recall of words. It is a modification of the Rey Auditory Verbal Learning Test (Lezak, 1995). The test consisted of oral presentation of 15 semantically unrelated words that subjects were requested to learn and recall in four consecutive trials. The score for total immediate recall was the sum of correctly recalled words. The score for delayed memory was the number of words recalled after 90 min and the score for delayed recognition was the number of correctly recognized words (15 targets and 15 distractors).

General intellectual ability (Full Scale Intelligence Quotient, FIQ) was estimated on the basis of six subtests of the Wechsler Adult Intelligence Scale (Wechsler, 1955).

The visual assessment of routine 21-channel EEG was performed to evaluate the background activity, focal (epileptiform/slow wave) as well as generalized (epileptiform/slow wave) abnormalities. The same neurophysiologist (E.M.), without knowledge of the clinical details, scored the EEG using criteria presented by Engel (1984).

2.3. Statistical analysis

The data were analyzed by using SPSS/PC + . Continuous variables are presented as means $\pm$ standard deviation. The results for categorized variables are presented as frequency estimates. The differences between the three groups (Group I, Group II and the controls) in memory measures were analyzed by covariance analyses (ANCOVA) with FIQ and age as covariates. Pairwise comparisons were performed by one-way analysis of variance with Duncan's post hoc test. Fisher's Exact test for independent observations was used for comparison of frequencies. $P < 0.05$ was considered statistically significant.

Logistic regression analysis is suitable for analyzing predictors of a binary outcome variable and consideration of multiple predictors. The following clinical variables were included in the analyses of predictive factors of 2-year seizure outcome: age at the time of the diagnosis; etiology of epilepsy, classified as remote symptomatic or cryptogenic; seizure type, subjects were categorized as having experienced either generalized seizures alone, or having experienced partial complex or mixed seizure types before the diagnosis; and pretreatment number of seizures classified into two categories, 10 or less and more than 10. From the EEG data, the presence or absence of spike focus and the presence or absence of paroxysmal epileptiform EEG activity were used. Neuropsychological variables were FIQ and verbal memory test scores. As the small size of the Group II sets limits for the number of predictors that could reasonably be entered into a multivariate model, memory variables for the logistic re-

gression analysis were selected on the basis of univariate analyses between the three groups (Group I, Group II and the controls). A backward stepwise logistic regression analysis was performed for clinical, neuropsychological and EEG variables. Odds ratios (OR) with 95% limits of confidence intervals (95% CI) were calculated from logistic regression model. CI limits were calculated using the formula $CI = \exp(B \pm 1.96SE[B])$, where B is the B coefficient in the logistic model.

2.4. Approval of the Ethics Committee

Approval for the study was given by the Ethics Committee of the University of Kuopio, Finland, and informed oral consent was obtained from all the patients.

Table 1
Baseline clinical and EEG variables in the seizure outcome groups

Variables		Group I ($n = 79$)	Group II ($n = 10$)
<i>Etiology</i>			
Remote symptomatic etiology		19 (24%)	5 (50%)
	Pre- and perinatal	5 (6%)	2 (20%)
	Post-traumatic	5 (6%)	
	Vascular	4 (5%)	2 (20%)
	Intracranial infection	3 (4%)	
	Congenital	2 (3%)	
	Toxic		1 (10%)
<i>Seizure history</i>			
Partial complex/mixed seizure type		18 (23%)	4 (40%)
Over 10 pretreatment seizures		11 (14%)	4 (40%)
<i>Interictal EEG</i>			
Paroxysmal epileptiform activity		9 (11%)	1 (10%)
<i>Focal activity</i>			
Spike focus		5 (6%)	4 (40%) ^a
	Temporal	5 (6%)	3 (30%) ^b
	Left	3 (4%)	3 (30%) ^b
	Right	2 (3%)	
	Extratemporal		1 (1%)
Only slow-wave focus		22 (28%)	3 (30%)
	Temporal	18 (23%)	3 (30%)
	Left	14 (18%)	1 (10%)
	Right	4 (5%)	2 (20%)
	Extratemporal	4 (5%)	

^a Fisher's Exact Test, $P < 0.01$.

^b Fisher's Exact Test, $P < 0.05$.

3. Results

3.1. 2-Year seizure outcome

Seventy-nine patients (89%) had satisfactorily controlled epilepsy (Group I), and 46 (58%) of these patients were seizure free after 2 years of antiepileptic drug (AED) treatment. Ten patients (11%) had refractory seizures (Group II).

3.2. Clinical and memory variables at baseline

The clinical and EEG variables are presented in Table 1. At the time of diagnosis, only the presence of spike focus in EEG differed significantly between the patients with satisfactorily controlled seizures and patients with refractory seizures ($P < 0.01$). When analyzed by the side of temporal lobe spike focus, the refractory patients more often

Table 2

Baseline demographic and neuropsychological variables in the seizure outcome groups and in the control group

Variables	Group I (<i>n</i> = 79)	Group II (<i>n</i> = 10)	Controls (<i>n</i> = 48)
Age	34.4 ± 15.3	29.2 ± 12.4	30.6 ± 11.8
Education	10.7 ± 2.8	10.6 ± 2.4	11.4 ± 2.3
Male/female	38/41	5/5	23/25
<i>Neuropsychology</i>			
Full-scale intelligence quotient	106.9 ± 13.4	105.6 ± 11.8	110.5 ± 11.2
Immediate story recall	11.5 ± 4.0	12.3 ± 4.7	13.2 ± 4.1
Delayed story recall	9.7 ± 4.3	10.6 ± 4.9	11.3 ± 4.8
Immediate list recall	29.9 ± 7.3	29.6 ± 8.3	34.2 ± 6.1 ^a
Delayed list recall	4.5 ± 3.3	3.2 ± 2.4	7.0 ± 2.5 ^b
Delayed list recognition	11.7 ± 2.7	9.9 ± 2.7	12.7 ± 2.0 ^c

^a ANCOVA, $P < 0.05$, Group II versus Controls.^b ANCOVA, $P < 0.001$, Controls versus Group I and Group II.^c ANCOVA, $P < 0.05$, Group II versus Group I and Controls.

had left-side focus than the satisfactorily controlled patients ($P < 0.05$), while there was no difference between the groups in the presence of right-side focus.

The patients and the controls did not differ regarding the demographic variables (Table 2). In the analysis of the memory measures, the ANCOVA with age and FIQ as covariates revealed significant group effect in total immediate recall of the words ($F = 3.66$, d.f. = 2, 131, $P < 0.05$), in delayed recall of the list ($F = 11.03$, d.f. = 2, 131, $P < 0.001$) and in delayed list recognition ($F = 4.18$, d.f. = 2, 131, $P < 0.05$) when the three groups were compared. Post-hoc analysis showed that the patients with refractory seizures learned less words than the controls ($P < 0.05$), the controls recalled more words after delay than both patient groups ($P < 0.05$), and the patients with refractory seizures performed worse in delayed recognition of the word list than the other two groups ($P < 0.05$).

The frequency of memory impairment was analyzed on an individual patient basis. The patients were classified as having clinically significant memory impairment when memory test scores were falling two standard deviations below the mean performance of the control group in at least two memory tasks. The number of patients with memory impairment at the time of diagnosis of epilepsy was, in Group I, 18 (23%) and in Group

II, five (50%). The difference between the groups was not statistically significant.

3.3. Predictors of 2-year seizure outcome

Five clinical and EEG variables, identified from the literature, three verbal memory measures, identified by univariate analyses between the groups, as well as age and FIQ were included in logistic regression analysis. In each of the three memory measures, the patients were classified according to presence of clinically significant impairment in verbal memory performance (scores < 2 S.D. from the control mean). Six risk factors were found to be significant predictors of refractory epilepsy in follow-up: younger age at the time of diagnosis, moderately impaired performance in immediate recall and in delayed recognition of the word list, presence of spike focus in EEG, remote symptomatic etiology of epilepsy, and partial complex or mixed seizure type before the diagnosis (Table 3).

The logistic regression model predicted the correct result in accordance with the observed 2-year seizure outcome in 94% of the cases. However, the model was better in predicting 2-year outcome of satisfactorily controlled epilepsy (100% correct) than unacceptable seizure control (50% correct).

The contribution of verbal memory impairment in predicting 2-year seizure outcome was estimated with a further logistic regression analysis,

which was performed with age, presence of spike focus in EEG, etiology of epilepsy and seizure type. With these predictors, 99% of the patients with satisfactorily controlled epilepsy were correctly classified. Only 20% of the patients with refractory seizures were correctly classified when the verbal memory variables were not included in the analysis.

As verbal memory impairment has been shown to be related to left temporal lobe epilepsy (Hermann et al., 1987), the contribution of left temporal spike focus to the seizure outcome was studied by performing an additional logistic regression analysis with the other original variables, except presence of spike focus, which was replaced by presence of left temporal spike focus in the analysis. The significant independent predictors of refractory seizure disorder were both the presence of left temporal spike focus (OR 15.4, $P = 0.02$) and impairment in delayed list recognition (OR 7.8, $P = 0.03$). The majority (99%) of the patients with satisfactorily controlled epilepsy and 40% of the patients with refractory seizures were correctly classified with the model. The model included also age at the time of diagnosis, remote symptomatic etiology of epilepsy, and partial complex or mixed seizure type, even though these variables did not attain statistical significance.

The ability of the six clinical and memory variables to predict seizure freedom 2 years after the diagnosis was analyzed in a logistic regression analysis with either complete ($N = 46$) or incomplete seizure control ($N = 43$) as the outcome variable. None of the predictors reached statistical significance. Sixty percent of the patients were correctly classified with the predictors, 63% of the

patients with complete seizure control and 56% of the patients with incomplete seizure control.

4. Discussion

The results of this study suggest that verbal memory impairment can be used as an early interictal predictor of seizure outcome in epilepsy. Our results demonstrate a multivariate model with clinical, memory and EEG variables that can predict a correct 2-year seizure outcome in 94% of newly diagnosed patients. However, with these factors, the predictions of satisfactorily controlled epilepsy could be made more accurately than predictions of refractory seizures or complete seizure freedom.

With the multivariate logistic regression analysis, we found six risk factors to be significant predictors of poor 2-year seizure outcome: younger age at the time of diagnosis, moderately impaired performance in immediate recall and in delayed recognition of the word list, presence of spike focus in EEG, remote symptomatic etiology of epilepsy, and partial complex or mixed seizure type before the diagnosis. The major predictor of refractory seizures was the presence of spike focus in the baseline EEG. In earlier studies of newly diagnosed epilepsy, the presence of epileptic, focal, slow-wave or background abnormalities in the pretreatment EEG have not been of prognostic value (Elwes et al., 1984; Collaborative Group for the Study of Epilepsy, 1992). However, in a recent study of risk factors of seizure recurrence after a first unprovoked seizure, focal epileptiform activity in the initial EEG was shown to be a significant predictor (Schreiner and Pohlmann-

Table 3
Predictors of refractory seizure disorder after logistic regression analysis

Predictor	<i>B</i>	<i>P</i> value	Odds ratio	95% CI ^a
Age at diagnosis	−0.11	0.02	0.89	0.82–0.98
Immediate list recall	2.49	0.04	12.1	1.06–136
Delayed list recognition	2.52	0.02	12.5	1.56–99.7
Spike focus	4.57	0.004	97.0	4.39–2140
Etiology	2.30	0.04	10.0	1.16–86.3
Partial complex/mixed seizure type	2.77	0.02	16.0	1.52–168

^a CI, confidence intervals of odds ratios.

Eden, 1997). In concordance with earlier studies (Elwes and Reynolds, 1991; Collaborative Group for the Study of Epilepsy, 1992; Mattson et al., 1996), partial complex or mixed seizure types before the diagnosis and remote symptomatic etiology of epilepsy were clinical predictors of refractory seizure disorder.

Age at onset of seizures has not been shown in previous studies to have prognostic value (Elwes et al., 1984; Collaborative Group for the Study of Epilepsy, 1992). However, age at onset of seizure disorder during adolescence has been shown to be a predictor of poorer outcome (Camfield et al., 1993; Braathen and Melander, 1997). In our study of newly diagnosed adult patients, including adolescents, younger age at the time of diagnosis was a predictor of poorer seizure control.

The patients with satisfactorily controlled epilepsy could be correctly classified according to the predictive model more frequently than the patients with refractory seizures. Thus, the absence of the six risk factors would predict a more favorable seizure outcome. Verbal memory impairment had more importance in predicting refractory seizure disorder than satisfactorily controlled epilepsy. In the refractory epilepsy group, only 20% of the patients could be correctly classified when the memory variables were not included in the model, but nearly all patients with satisfactorily controlled epilepsy could be correctly classified with clinical predictors alone. Thus, evaluation of memory function can provide further predictive information when other clinical risk factors of intractable epilepsy are present at the time of the diagnosis of partial epilepsy.

Left temporal spike focus and impairment in verbal memory are commonly found in temporal lobe epilepsy (Hermann et al., 1987; Perrine et al., 1991). In this study, both verbal memory impairment and left temporal spike focus were significant risk factors of refractory epilepsy, and other clinical risk factors did not reach statistical significance. This suggests that left temporal lobe epilepsy with memory impairment has poorer prognosis. However, the possible hippocampal pathology could not be analyzed because the magnetic resonance imaging (MRI) data was not available from all the patients. The contribution

of right temporal spike focus to the prognosis could not be analysed because there were no patients with a right temporal focus in the refractory epilepsy group.

The predictive factors of the model were not as efficient in predicting complete seizure freedom, which is the ultimate goal of the treatment of epileptic seizures. It is possible that patients with refractory epilepsy as a group are distinct from patients with satisfactorily controlled epilepsy and determinants of their seizure outcome may be partly different.

In recent years, multivariate models have been applied to identify predictors of seizure outcome after epilepsy surgery (Spencer et al., 1993; Guldvog et al., 1994; Armon et al., 1996) but the results of the predictive factors have been inconsistent. Neuropsychological variables have been used in a multidisciplinary study of predictors for epilepsy surgery outcome, and measures of cognitive abilities and psychosocial adjustment were found to predict seizure relief (Dodrill et al., 1986). The contribution of neuropsychological data alone to the prediction of postoperative seizure control has also been studied (Sawrie et al., 1998). The results of the studies suggest that neuropsychological data used in a multivariate statistical fashion may offer an incremental increase in prediction of seizure outcome after epilepsy surgery.

Memory performance is of importance in epilepsy because of the prominence of the temporal lobes in epileptogenesis and in memory processing (Jones-Gotman et al., 1993). Memory deficits have been documented in intractable epilepsy (Hermann et al., 1987), but also newly diagnosed patients with cryptogenic partial epilepsy have been shown to have at least mild verbal memory dysfunction (Äikiä et al., 1995). In the present study, the patients with satisfactorily controlled seizures had impaired delayed recall compared with healthy controls, but the patients with refractory seizures had initially impaired performance in both immediate and delayed verbal memory. The difference between the two patient groups was seen in the delayed recognition task, in which the patients with refractory seizures had poorer performance. However, the patient groups

were comparable when memory impairment was defined according to clinically significant criteria (two standard deviations from the mean performance of the controls) and analysed on an individual patient basis. These results can be summarized as suggesting that in group comparison studies, impaired memory performance is related to poor seizure outcome, but if memory impairment is analysed on an individual patient basis and with clinically meaningful impairment criteria, the relationship is not as evident. This reflects the general notion that in neuropsychological assessment, group effects may be present but it is often more difficult to apply these results on an individual patient basis (Loring, 1997).

Severity of hippocampal damage measured either by volumetric MRI or by histologic analysis has been shown to correlate with the impairment of memory performance in chronic epilepsy (Sass et al., 1990; Lencz et al., 1992). Also, in newly diagnosed patients with left temporal lobe epilepsy, MRI-based measurements of left hippocampus have been shown to correlate with verbal memory performance (Kälviäinen et al., 1997). Recently, it has been suggested that the etiology of the hippocampal sclerosis is an acquired injury to the hippocampus and the type of the initial injury influences the course and outcome in epilepsy (Mathern et al., 1995). In addition, recurrent seizures over many years have been shown to be the determinant for reduction in hippocampal volume, particularly when associated with the early onset of epilepsy (Kälviäinen et al., 1998; Salmenperä et al., 1998). It is possible that the presence of early structural and functional deficits in epilepsy affect the prognosis. In multivariate analyses, the use of combination of MRI-based measurements and memory variables as well as EEG and clinical data could increase the validity of the predictive model of seizure outcome in newly diagnosed epilepsy.

Our results of verbal memory impairment as a predictor of seizure outcome can be regarded as preliminary because of the relatively small number of patients, especially in the refractory epilepsy group. This causes imprecision to the results as seen in wide confidence intervals of odds ratios for the prognostic factors. The number of

patients was also too small to perform a cross-validation of the model. However, the need for new prognostic factors of intractable epilepsy is evident, and evaluation of memory function can provide additional information when clinical risk factors of refractory epilepsy are present at the time of the diagnosis of partial epilepsy. The patients whose seizures will prove to be refractory should be identified as early as possible in order to improve their prognosis with best possible interventions including rational antiepileptic drug therapy, surgical treatment and new innovative treatment options.

Acknowledgements

We gratefully acknowledge Pirjo Halonen, at the Computing Center of the University of Kuopio, for her significant contribution in statistical analyses.

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